Compositional Neural Mechanics with Observable Physical Uncertainty

Working research manuscript | Simulation study | 2 November 2026

Abstract

A reconstruction can reproduce observed motion while predicting the wrong response to a new interaction. We study a representation that separates reusable bulk material energies from relational attachment energies, and composes both through a nonlinear equilibrium solver. Scene-specific continuous descriptors encode material heterogeneity, support stiffness, nonlinearity, location and extent. Prescribed tool positions and orientations are inputs; deformation and reaction forces are outputs. We distinguish forward-law transfer with known descriptors from inference with unknown descriptors, and distinguish numerical convergence from uncertainty calibration.

The implementation preserves an exact common-stiffness scaling symmetry: with zero body load, scaling all stiffnesses changes forces but leaves every position-controlled equilibrium unchanged. Known gravity and density can supply the missing load reference. Development experiments demonstrate accurate nonlinear response, improved inverse optimization, and useful refinement from an additional observation. A reserved composition and geometry evaluation and an independent numerical posterior check are reported below when their audits finish. The present scope is quasistatic tracked-point mechanics with known geometry and categorical law families; it does not establish a complete video-to-simulator system or superiority to published vision systems.

1. Scientific question

The target is counterfactual physical accuracy: given an observation history, can a model predict the deformation and reactions under a previously unseen command? Rendering agreement measures a different quantity. Multiple physical explanations can produce the same visible shape, and a better image fit need not reduce error in future forces. We therefore report physical outputs separately and evaluate observations through their effect on future predictions.

Contribution under test

We test whether shared local laws, composed by the same mechanical operator, transfer across material-support combinations and geometry more systematically than direct displacement prediction. A second question is whether scene inference and action-conditioned refinement preserve physical ambiguity while improving useful predictions. Neural constitutive laws, differentiable mechanics, Gaussian support fields and active sensing are established ingredients; the proposed contribution is their composition and intervention-based validation.

2. Representation and mechanics

Material coordinates X remain attached to the object. Its state is x(X)=X+u(X), with deformation gradient F=I+grad u. Geometry is a mapped cuboid with optional taper and arch. The present material field is isotropic and hyperelastic; spatial variation is expressed in material coordinates rather than a fixed world-space field.

```
W(F, [s,z]) = exp(s) W_shared(F,z)
P = dW/dF
```

The bulk network uses rotational invariants and positive coefficients. Factoring out the common stiffness exp(s) enforces the scaling identity exactly. This construction provides objectivity and a consistent energy-stress relation; we do not claim a general polyconvexity guarantee. The training families are Mooney-Rivlin and Yeoh-like responses, with continuous scale and shape descriptors.

A separate shared attachment energy depends on relative displacement between a material point and its commanded tool target. Gaussian footprints specify location and width. The reference-area-normalized energy includes positive stiffness and nonlinear shape parameters, and supports polynomial and saturating response families. These are bilateral compliant attachments, not nonpenetrating contact constraints.

```
Pi(u) = integral W(F,z_m) dV
+ sum integral psi(delta(u,a),z_b) dA
- integral rho g . u dV
R(u,z,a) = dPi/du = 0
```

A quadratic finite-element discretization, Newton iterations and line search solve equilibrium. The implementation checks residuals and positive deformation Jacobians. Gravity and density remain explicit body-load inputs. Forces and torques on the prescribed tools are derived from the same energy and solved state.

Scene inference

The 15 continuous unknowns are bulk log stiffness, shape and heterogeneity amplitude; two attachment log stiffnesses and nonlinearities; four footprint center coordinates; and four log widths. Geometry, categorical bulk/interface family and heterogeneity pattern are conditioned on. The learned law weights stay fixed while these scene descriptors are fitted to noisy tracked surface positions. Reaction observations enter only in explicitly labeled measurement variants.

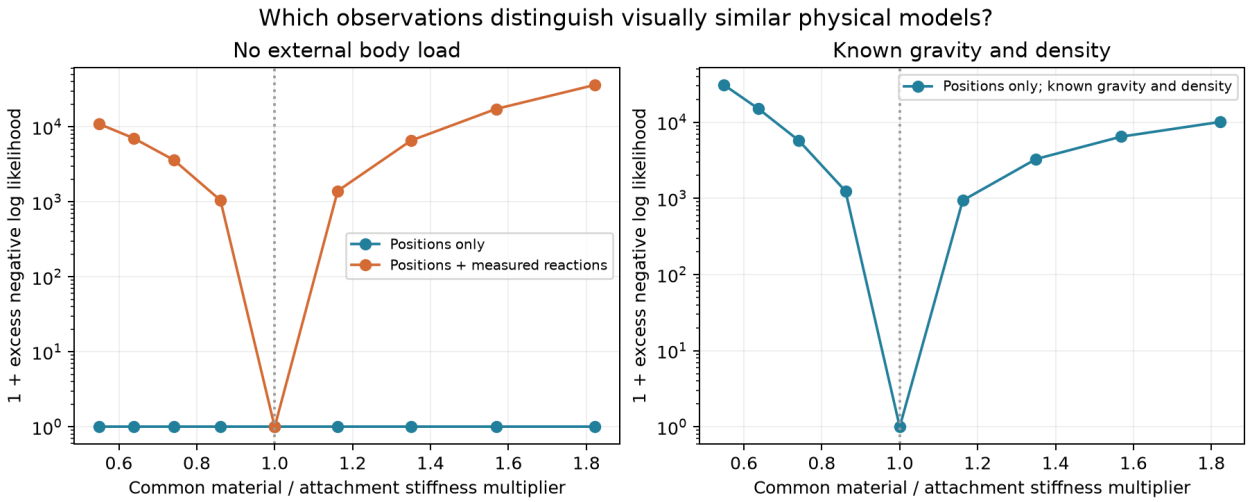
```
H du/dz = -dR/dz
L(z) = 0.5 ||r_observation(z)||^2 + 0.5 ||r_prior(z)||^2
```

Implicit sensitivities differentiate the converged mechanical state. A bounded Gauss-Newton trust-region solver uses the full residual Jacobian and its scaling to improve optimization conditioning. Its stopping status is reported separately from parameter accuracy, predictive accuracy and posterior validity.

3. Observability and refinement

Common-scale ambiguity. If gravity and other prescribed body forces vanish, multiplying every bulk and attachment stiffness by c multiplies the entire energy by c. The equilibrium displacement is unchanged, whereas the reactions multiply by c. All rendered views of that state are therefore unchanged. This ambiguity persists across every prescribed-position experiment in this model; additional cameras cannot remove it.

Known gravity with known density introduces a fixed reference load and changes this conclusion: stiffness affects visible sag. If density is also free to scale with stiffness, a joint ambiguity remains. These assumptions matter when claiming that position-only observations identify physical properties.



Development likelihood slices at fixed relative stiffnesses. These are not marginalized posteriors or confidence intervals. The zero-gravity and known-gravity experiments use different noise seeds.

Functional uncertainty

A local Gaussian approximation uses the observation Jacobian and an explicit prior precision. Candidate actions are ranked by their expected reduction in variance of a specified future response, rather than total parameter variance. The target commands are public inputs; their response labels are unavailable during selection.

```
C = (J_obs^T J_obs + P_prior)^(-1)
C_after(a) = (C^(-1) + J_a^T J_a)^(-1)
score(a) = trace[A (C - C_after(a)) A^T]
```

This approximation does not automatically account for nonlinear multimodality, active bounds or model discrepancy. The development bands set model discrepancy to zero and are labeled nominal. A separate small reference posterior tests integration convergence before any claim that its intervals are trustworthy. Empirical coverage and fixed/random acquisition controls remain required.

4. Development evidence

Forward model	Position RMSE (mm)	Force RMSE (N)	Torque RMSE (Nm)
Shared laws	0.0190	0.00704	0.000099
Neo-Hookean	0.0494	0.04496	0.000252

Four validation objects, six commands each; both models use known descriptors and n=4. One original runtime failure was replayed in an isolated process with identical settings; the failure record is retained. These validation numbers are not the reserved test.

Object / model	Original position (mm)	Gauss-Newton (mm)	GN force (N)
0 / analytic	0.0092	0.0088	0.00245
0 / learned	0.0101	0.0091	0.00345
1 / analytic	0.0243	0.0190	0.00256
1 / learned	0.0250	0.0219	0.00225

Full 15-parameter development inversion, two observed objects and two future commands each. All four Gauss-Newton fits stopped successfully by ftol, versus zero original fits within their budgets. All motion errors decreased; force errors increased for object 0 and decreased for object 1. This is an optimization comparison, not an independent generalization result.

Observed failure modes and repairs

A narrow-slip attachment training distribution produced a 4.065 mm same-mesh error in an exposed stretching case. A separately documented warm-start, wide-slip training run reduced that case to 0.079 mm. A failed alternative wide-range run is retained. The repaired law was frozen before reserved testing; the exposed case is not counted as evidence of unseen-scene transfer.

Development mesh checks also showed that n=4 was inadequate for several extreme reference cases; targeted n=6 to n=8 checks passed with field differences of 2.98-8.15 micrometres. The reserved benchmark consequently uses a fixed n=8 analytic reference. That choice is supported by development tests, not individual convergence certification for every reserved scene.

5. Reserved forward transfer

Training uses 54 objects from 18 geometry-material-interface-heterogeneity cells and six commands, plus four validation objects. Six factor combinations are withheld while each pair of factors remains represented. The reserved test has six withheld-composition objects and six draws on two new geometries, each evaluated under four new command directions or combinations. Test identities, checkpoints, source hashes and methods are committed before generating results.

All forward methods receive oracle descriptors. Comparisons include the matched analytic family at n=4, small-strain-matched Neo-Hookean, learned laws at n=4, and custom direct point predictors trained for 6,000 and 30,000 steps. The direct predictors receive numeric law fingerprints from the same learned laws; this provides a shared physical prior but is not computationally identical to full solver access. These custom networks are not reproductions of PhysTwin or MeshGraphNet.

Reserved results pending independent completion audit. No partial-cohort averages are reported.

Preprocessing control

A source-only review identified that training-constant command coordinates were divided by a standard deviation floor of 1e-8, causing extreme features for new directions. The primary frozen baselines are retained. A supplementary control replaces only exactly zero-mean, zero-variance input scales with fixed physical units; weights and all training predictions remain unchanged. Its predictions are committed before reading reference outputs. Final interpretation must include this control rather than crediting the mechanics architecture for a normalization defect.

6. Observation refinement and uncertainty checks

Policy	Mean position (mm)	Mean nominal 90% coverage
initial	0.3708	98.4%
fixed_stretch	0.0170	85.3%
fixed_right_bend	0.1119	92.0%
fixed_opposed_twist	0.0295	81.4%
random	0.0235	80.4%
target_task_variance	0.0170	85.3%

Two exposed development objects and two future commands per object. Each non-initial policy receives exactly one additional observation. The variance policy selected stretch for both objects and therefore ties fixed stretch. Pointwise coordinates are correlated; these coverage fractions are descriptive, not independent calibration trials or evidence of a superior policy.

Reference posterior

The three-parameter conditional reference posterior is not yet numerically qualified. No trustworthy interval claim is made from the current run.

This reference calculation exploits the exact zero-gravity common-scale identity: it solves in two relative stiffness coordinates and integrates the common scale separately with the original bounded prior. Grid refinement, conditional integration and proposal-width checks are required. It is a diagnostic of a small conditional inverse problem, not a 15-dimensional posterior validation.

A separately frozen six-object inverse study is running with unknown continuous descriptors, both categorical bulk and attachment families represented, n=4 analytic observations and n=2 inference. It compares fitted analytic and learned laws, with fixed, random and task-variance policies at equal observation budgets. The discretization mismatch intentionally tests whether the same-mesh development result survives forward-model error.

Contact extension

A separate normal-penalty operator evaluates signed distance to prescribed planes and spheres and returns consistent rigid reactions. Plane contact passed the development mesh and quadrature checks. Sphere and combined cases still showed mesh-sensitive fields or reactions, and a tiny contact patch was missed by coarse quadrature. These failures remain explicit. Finite penetration, absent friction and prescribed rigid geometry prevent treating this extension as a general coupled-contact result.

7. Relation to vision-based physical reconstruction

PhysTwin [1] reconstructs deformable digital twins from sparse multiview videos using shape priors, spring-mass mechanics and Gaussian appearance. Our current tracked-point, known-geometry study isolates local-law composition and inference; it is not an end-to-end comparison with that reconstruction system.

PhysGS [2] maintains Bayesian physical-property beliefs over Gaussian representations and refines them from observations. Our evaluation instead targets future mechanical responses and explicit observability under a coupled equilibrium model. Bayesian property estimation itself is not a novel ingredient here.

PhysHO [3] incorporates neural constitutive modeling into human-object interaction simulation, while **NeuSpring [4]** uses neural spring-property fields. These works preclude claiming that a neural material field or constitutive component alone is new. Our hypothesis concerns reusable bulk/interface composition and its validation under unknown scene descriptors and new interventions.

8. Scope and next evidence

The implementation supports useful pieces of the proposed architecture: reusable bulk and attachment laws, position-controlled equilibrium, scene inference, structural observability and task-aware action scoring. Current evidence is restricted to mapped cuboids, known categorical families, known geometry and quasistatic elasticity. It does not establish arbitrary unseen material families, topology changes, soft-soft contact, friction, impacts, learned perception, Gaussian rendering quality or new-view acquisition.

The next decisive evidence is a separately untouched inverse cohort, with all continuous material/support descriptors hidden, matched observation budgets and strong fitted analytic and fixed-action controls. Reference discretization error and learned-model discrepancy must be separated from latent uncertainty. Existing repository benchmark results remain preserved but cannot be attributed to this new representation without adapting and rerunning it.

References

[1] H. Jiang et al. PhysTwin: Physics-Informed Reconstruction and Simulation of Deformable Objects from Videos. ICCV 2025. Primary source.

[2] S. Chopra et al. PhysGS: Bayesian-Inferred Gaussian Splatting for Physical Property Estimation. CVPR 2026. Primary source.

[3] S. Jiang and G. H. Lee. PhysHO. CVPR 2026. Project and paper. Primary source.

[4] Q. Xu et al. NeuSpring. arXiv:2511.08310, 2025. Primary source.